GLA University, Mathura
II-Mid Term Examination, 2011-12

Course: - B.Tech. **II-Year, IV-Sem.**

Subject: - Computer Graphics (CSE 223) **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-20**

Notes:-

2) Answer all questions, Section A is **compulsory, Any Two** from Section B and **Any Two** from Section C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

PART-A

Objective Questions: (0.5*16 = 08)

I. A pure rotation means that the determinant of T-matrix is

A. 0 B. 1 C. -1 D. 2

II. If $S_x=S_y=0$ then it means:

A. Uniform reduction B. Object collapses C. Uniform scaling D. None of above

III. The transformation occurring in 2D- graphics are:

A. Circular B. Linear C. Non-linear D. All above

IV. Several transformation matrices can be stored in a data structure called as:

A. Queue B. Stack C. linked list D. Circular queue

V. A technique of transition of one shape into other is known as:

A. Animation B. Morphing C. Transformation D. Projection

VI . The join points of the curve are also known as:

A. Plots B. Knots C. Clots D. Dots

VII. Fractals stands for:

A. Fractional part B Fractional Dimensional C. Fractional point D. All of these

VIII. A parameter that controls how tightly or loosely the spline fits the control graph is:

A. Bias Parameter B. Tension Paramenter C. h- parameter D. None of these

IX . In perspective projection, the center of projection is at:

A. Infinity B. Finite Distance C. x-axis D. y-axis

X . If the view plane is moved away from the reference point, the projected size goes on:

A. Decreasing B. Increasing C. To be same D. None of these

Fill in the blanks Questions:

XI. In parallel projection, the center of projection is at _____.

XII. The plane on which we are taking the projection is called as_____.

True and False Questions:

XIII. The steps for Rotate an object about an arbitrary point (h,k): **T-vRθTv**

XIV. Rotation is a transformation that produce mirror image of an object .

XV. Spline approximation method was developed by Catmull.

XVI. If the projection plane normal makes unequal angles with each principal axis then, the projection is called Isometric

PART-B

I. What is Projection? Differentiate Parallel and Perspective projection with example. (2)

II. What is Transformation? Explain the need of Transformation with proper example and inverse transformation. (2)

III. A Cubic curve is defined by the points (1,1),(2,3),(4,4) and (6,1). Calculate the coordinates of parametric midpoint of the curve and verify that its gradient dy/dx is 1/7 at this point. (2)

PART-C

I. Perform a 45^0 rotation of triangle A(0,0) B(1,1), C(5,2), About origin and about point P(-1,-1). (4)

II. Reflect a triangle whose vertices are A(-1,0), B(0,-2),C(1,0)about the line y=x+2. (4)

III. Explain Bezier curves with example. Write the properties of B-Spline curves and Bezier curves. (4)

GLA University, Mathura

II-Mid Term Examination, 2011-12

Course: - B.Tech (CSE) II-Year, IV-Sem.

Subject: -Computer networks **Subject code-CSE224** **Uni. Roll No:-**

Time:-90 Minutes **Max. Marks:-20**

Notes:-

1) Answer all questions, Part A is **compulsory, Any Two** from Part B and **Any Two** from Part C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

Part-A

I. Attempt all questions. **8**

Q1: Choose the correct answer for each of the following. **(1/2*12)**

i) Which error detection method involves polynomials?

a) Simple parity check b) Two-dimensional parity check c) CRC d) Check-sum

ii) Which error detection method uses ones complement arithmetic?

a) CRC b) Hamming Code c) Simple Parity Check d) Check-Sum

iii) Flow control is needed to prevent _____.

a) bit errors b) overflow of the sender buffer c) overflow of the receiver buffer d) collision between sender and receiver

iv) An example of piggybacking is when.

a) packets cross paths b) acknowledgment packet is sent with a data frame

c) packets collide and must be recovered or "piggybacked" d) All of the above.

v) For Stop-and-Wait ARQ, for N data packets sent, _____ acknowledgments are needed.

a) N-1 b) 1 c) 2N d) N+1

vi) Which of the following devices is not used to regenerate or refreshen signal strength?

a) Bridge b) Hub c) Repeater d) all of them

vii) A good polynomial generator needs to have the following characteristic.

a) It should have at least two terms. b) It should have the factor x + 1.

c) The coefficient of the term x^0 should be 1. d) All of above

viii) In a cyclic code, If $s(x) = 0$, either

a) No bit is corrupted b) Some bits are corrupted, but the decoder failed to detect them

c) a or b d) none of the above.

ix) If minimum Hamming distance of a coding scheme is 4, receiver can detect-----and correct--- bit error.

a) 3, 2 b) 3, 1 c) 3, 3 d) 1, 1

x) A bit string, 0111101111101111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing?

a) 011110111110011111010 b) 011110111100011111010

c) 011110111110011111110 d) 0111101111101111110

xi) If the header of the frame allows m bits for the sequence number, in Go-Back-N ARQ, the size of send window and receive window are.

a) 2^m and 1 b) $2^{m/2}$ and $2^{m/2}$ c) $2^{m/2}$ and 1 d) none of the above.

xii) The stations on a wireless ALOHA network is a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, assume that sender has transmitted a frame and acknowledgement is not received then Back-off time may be.

a) 1 m.s. b) 2 m.s. c) 3 m.s. d) 4 m.s.

Q2: State True/False **(1/2*4)**

i. The Three common redundancy methods are: parity check, cyclic redundancy checks (CRC) and zero-sum.

ii. If G is the average number of frame generated by the system during one frame transmission time. The throughput for slotted ALOHA is $S = G \times e^{-2G}$.

iii. The high-speed Token Ring networks called FDDI (Fiber Distributed Data Interface) and CDDI (Copper Distributed Data Interface) use dual ring topology.

iv. A repeater is a connecting device that operates at the data-link layer of the Internet (TCP/IP) model.

Part-B

II. Attempt any two **(2*2)**

Q1: Differentiate between the following:

i. Pure ALOHA and Slotted ALOHA

ii. Bridge and Layer 2 Switch

Q2: Consider building a CSMA/CD network running at 100 Mbps over a 2-km cable with no repeaters. The signal speed in the cable is 200,000 km/sec. What is the minimum frame size?

Q3: Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose propagation time from the earth is 270 msec. Acknowledgements are always piggybacked onto data frames. The headers are very short. Four-bit sequence numbers are used. What is the maximum achievable channel utilization for Selective repeat ARQ?

Part-C

III. Attempt any two **(4*2)**

Q1: Answer the following

a) How the senders can sends 4bits (0110) using Hamming code? Detect and correct the error if the receiver receives bit pattern (0110111) sent using a Hamming code?

b) A block of address is granted to a small organization. We know that one of the addresses is 205.16.37.39/28. Find-

- First address of organization
- Last address of organization

c) Explain address depletion in classful addressing scheme?

Q2: What is channelization? Explain the channelization protocols in brief?

Q3: Write short notes on following-

- Polling
- IEEE standard for LANs

GLA University, Mathura
Second Mid-Term Examination, 2011-12
Course: B.Tech. (CSE), II-Year, IV-Sem.
Database Management Systems (ITE221)

Uni. Roll No:

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question (a, b, etc.) should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Q1. Attempt all. **.5x16=8**

I. Which of the following is an advantage of view?
a. Data security b. Derived columns c. Hiding of complex queries d. All of the above

II. Which of the following is a comparison operator in SQL?
a. = b. LIKE c. BETWEEN d. All of the above

III. Which of the following database object does not physically exist?
a. table b. index c. view d. none of the above

IV. A table joined with itself is called
a. Join b. Self Join c. Outer Join d. Equi Join

V. The NVL function
a. Assists in the distribution of output across multiple columns.
b. Allows the user to specify alternate output for non-null column values.
c. Allows the user to specify alternate output for null column values.
d. Nullifies the value of the column output

VI. Which of the following SQL statement is used to remove a database object, such as table, from the database?
a. REMOVE b. DROP c. KILL d. DELETE

VII. Normalization is a process of restructuring a relation to
(a) minimize duplication of data in a database
(b) maximize duplication of data to ensure reliability
(c) make it of uniform size
(d) allow addition of data

VIII. Given an attribute x, another attribute y is dependent on it, if for a given x
(a) there are many y values (b) there is only one value of y
(c) there is one or more y values (d) there is none or one y value

IX. Fourth normal form (4 NF) relations are needed when
(a) there are multivalued dependencies between attributes in composite key
(b) there are more than one composite key
(c) there are two or more overlapping composite keys
(d) there are multivalued dependency between non-key attributes

X. The essential characteristic of ..... Normal form is that every determinant in the table must be a candidate key.
a. 3NF b. BCNF c. Forth NF d. Fifth NF

XI. The 2NF describe that:
a. There are no repeating groups in the table

PTO...

b. All attributes are dependent on the primary key
c. a & b with no partial dependency
d. there is no partial dependency

XII. Use the ..... Clause to restrict groups returned by a grouped query.
a. Where b. Distinct c. Having d. Uniquerows

XIII. What type of join is needed when you wish to include rows that do not have matching values?
a. Equi-join b. Natural join c. Outer join d. All of the above.

XIV. Which of the following is a correlated subquery?
a. Uses the result of an inner query to determine the processing of an outer query.
b. Uses the result of an outer query to determine the processing of an inner query.
c. Uses the result of an inner query to determine the processing of an inner query.
d. Uses the result of an outer query to determine the processing of an outer query.

XV. The different classes of relations created by the technique for preventing modification anomalies are called:
a. normal forms. b. referential integrity constraints.
c. functional dependencies. d. None of the above is correct.

XVI. A functional dependency is a relationship between or among:
a. tables. b. rows. c. relations. d. attributes.

Q2. Attempt any two. **2x2=4**

A. Explain the different anomalies in database with example.
B. Given relation R(A, B, C, G, H, I) with dependencies
F= {A → B, A → C, CG → H, CG → I, B → H}
Is AG a candidate key?
C. Discuss the SQL and its functions.

Q3. Attempt any two. **4x2=8**

A. Discuss functional dependency with example? Also explain inference rules?
B. What are the desirable properties of decomposition of a relation? Explain Lossless-join decomposition. Given R(A,B,C,D,E) with the functional dependencies F{AB-> CD, A->E, C->D}. Check whether the decomposition of R into R1(A,B,C), R2(B,C,D) and R3(C,D,E) is lossy or Lossless.
C. Consider the following tables:
Dept(Deptno, Dname, Location)
Emp (Empno, Ename, Job, Mgr, Hiredate, Sal, Comm, Deptno)
MGR is the empno of the employee whom the employee reports to.
Write the following queries in SQL:
i. List the employees who are earning more than 12000 but less than 40000.
ii. List the employee names those are starting with 'J' and with five characters.
iii. List the employees who are joined in the year 81.
iv. List the Employees who are located at 'New York'.
v. List the department wise total salary.
vi. Display the employee name and their department name.
vii. Display the employee name whose salary is larger than the average salary of their department.
viii. Display the name of the employee getting highest salary.

B.Tech (CSE) II Year, IV Semester
Second Mid Term Examination – 2012
Introduction to Microprocessor-CSE222

Time: 1 Hr 30 Mins MM: 20

Notes:-

1) Answer all questions, **Part A** is **compulsory**, **Any two** from Part B and **Any two** from Part C.
2) All parts of a question (a, b, etc.) should be answered at one place.
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Part A

Q1) Attempt following questions. **[0.5x16=08]**

i) A Bus cycle is equal to how many clocking periods?
A) Two B) Three.
C) Four D) Six

ii) NMI input is
A) Edge sensitive B) Level sensitive
C) Both edge and level triggered. D) edge triggered and level sensitive

iii) What do the symbols [ ] indicate?
A) Direct addressing B) Register Addressing
C) Indirect addressing D) None of the above

iv) One thousand bytes represent a
A)Megabyte B)Gigabyte C)Kilobyte D)None of the above

v) In a DMA write operation the data is transferred
A) From I/O to memory. B) From memory to I/O.
C) From memory to memory. D) from I/O to I/O.

vi) A computer cannot "boot" if it does not have the?
A)Compiler B)Loader C)Operating system D)Assembler

vii) Which of the following is not hardware
A) Magnetic tape B) Printer C) VDU terminal D) Assembler

viii) Direction flag is used with
A) String instructions. B) Stack instructions.
C) Arithmetic instructions. D) Branch instructions

ix) Status register is also called as.
A) Accumulator B) Stack C) Counter D) flags

x) The Stack pointer holds
A) 16 bit address B) 16 bit data C) 8 bit address D) 8 bit data

xi) The communication line between the CPU, memory and peripherals is called a
A)Bus B) Line C) Media D) none of these

xii) Number of the times the instruction sequence below will loop before coming out of loop is

```
MOV AL, 00h
A1: INC AL
JNZ A1
```

A) 00 B) 01
C) 255 D) 256

xiii) 8086 microprocessor has

A) 16 bit data bus B) 20 bit address bus

C) 6 byte pre-fetch queue D) All of the above

xiv) Which microprocessor pins are used to request and acknowledge a DMA transfer?

A) reset and ready B) ready and wait

C) HOLD and HLDA D) None o these

xv) In which T-state does the CPU sends the address to memory or I/O and the ALE signal for demultiplexing

A) T1 B) T2

C) T3 D) T4

xvi) How many I/O ports can be generated by 8086?

A) 65536 B) 65365

C) 65845 D) 65865

Part B

Q2) Attempt any two of the following. **[2x2=04]**

a) Explain flags of 8086 microprocessor system?

b) What is subroutine? Discuss the advantages and disadvantages of subroutine?

c) Write instructions to a) load 00H to accumulator

b) Decrement the accumulator

c) Display the answer

Part C

Q3) Attempt any two of the following. **[4x2=08]**

a) Explain the concept of LOOP and Counter in assembly programming language with an example?

b) Write an ALP to add two 16-bit numbers? Calculate the time to execute the given program. If frequency of microprocessor is 2 MHz?

c) Draw the block diagram of 8086 and specify the function of different units of 8086 processor?

GLA University Mathura,

II Term

B.Tech: II Year, 4th semester [M.E., C.S&E.C.]

Ethics & Values (AHE: 201)

Marks: 20 Time: 1.30 Hours

There are three sections in this question paper. Section A consists of very short answer type questions, while section B & C consist of descriptive answer type questions. Attempt questions from all sections.

Section A

1) Write short notes on each of the following :(Word Limit: 30 words, Marks: 1 mark each)

a) Bhopal Gas
b) The case of Nestle (Baby Milk)
c) 51-L Challenger
d) Professional Responsibility
e) Self-interest
f) Shared Responsibility
g) Profession
h) MTI

Section B

2) Answer any two of the following :(Word Limit: 150 words, Marks: 2 marks each)

a) Explain the concept of an engineer as a professional.
b) What do you mean by the due-diligence?
c) What do you mean by the professional-identity?

Section C

3) Answer any two of the following :(Word Limit: 250 words, Marks: 4 marks each)

a) An engineer should not behave like a manager. Comment on this statement by analyzing the case of 51-L Challenger.
b) What do you mean by the ethics of self-interest? Analyze it through a case.
c) What do you mean by professional-autonomy? Analyze it through a case.